

A Label–Score Alignment Pitfall in SNR-Conditioned OOD Detection for Single-Channel Blind Source Separation

Bin Nong, Weihong Fu, and Zhuoyun Jiang

Abstract—Evaluating out-of-distribution (OOD) detection for communication signals is routinely conditioned on the operating SNR. We show this conditioning is fragile to label misalignment. Storing per-source SNR labels interleaved (`np.repeat`) while the scores they annotate are stacked slot-wise (`np.tile`) manufactures a spurious phenomenon in which two OOD scorer families appear complementary across SNR and a training-free SNR-routed ensemble appears to lift AUROC from 0.50 to 0.625 ± 0.031 . We formalize the failure as an indexing-permutation confound and prove that whenever OOD scores drift with SNR, *any* label permutation that alters the per-bin SNR composition converts that drift into spurious per-SNR separability; a closed-form Gaussian corollary and a domain-free toy show the confound needs no separator, signal, or detector. The same pipeline also hides a second, independent misalignment—per-source labels swapped alongside permutation-invariant alignment—with the opposite effect: a genuine same-mixture detection margin (AUROC 0.54) attenuated to an apparent null. Randomness-oriented safeguards (multi-seed averaging, significance tests, robustness batteries) detect neither. Doubly corrected, no post-hoc scorer gives deployable *pooled* discrimination (rounded 0.508 ± 0.026 ; five seeds, four test sets); a modest *same-mixture* embedding margin survives (prototype AUROC 0.54), inverted for logit scorers and collapsing when the unknown source is weak.

Index Terms—Blind source separation, open-set recognition, out-of-distribution detection, evaluation methodology, label misalignment.

I. INTRODUCTION

SINGLE-CHANNEL blind source separation (SC-BSS) of co-frequency communication signals is increasingly paired with per-source modulation identification [1], [2], and a deployed receiver must additionally *reject* sources whose modulation it has never seen—an open-set recognition problem [3], [4] for which post-hoc OOD scorers of learned embeddings and logits [5], [6], [7], [8], [9] are the standard toolkit. Because OOD score distributions drift with noise, communication-signal evaluation is routinely conditioned on the operating SNR [14], [15], [4].

This letter shows that the conditioning step itself is a single point of failure. A one-line labeling-convention mismatch misaligns every known-pool score with the wrong SNR bin, and each “per-SNR bin” then compares known and unknown pools at *different* true SNRs: the bins measure SNR separability, not

OOD separability. The failure is insidious precisely because it is *systematic*: multi-seed averaging, significance testing, and robustness batteries are randomness-oriented safeguards, and they cannot detect deterministic pipeline errors [11], [12]. We contribute: (i) a formalization of the failure as an indexing-permutation confound—a proposition and a closed-form Gaussian corollary giving the exact conditions (SNR-dependent scores plus a composition-changing label permutation) under which *any* such misalignment creates spurious per-SNR gains, verified analytically and by a domain-free toy, and why the trusted checks miss it; (ii) a deployment-faithful evaluation protocol (verified alignment, strict train/reference/test separation, unknown-free calibration, operating-point and same-mixture metrics, independent test realizations)—whose same-mixture control exposes a *second*, independent label misalignment in the same pipeline (per-source labels swapped alongside permutation-invariant alignment), with the opposite sign: it attenuates a genuine margin to an apparent null; and (iii) the doubly corrected study on an open-set SC-BSS benchmark (six scorers, five seeds, two backbones, four test sets, LOMO splits, an interference sweep): no deployable *pooled* discrimination, but a modest *same-mixture* embedding margin that is inverted for logit scorers and collapses when the unknown source is weak.

II. OPEN-SET SC-BSS SETUP

Signal model and SNR definition. We study the two-source co-frequency mixture $y(t) = \alpha_1 s_1(t) + \alpha_2 s_2(t) + n(t)$ at complex baseband. Each s_i is a unit-power synthetic waveform (16 kHz sampling, 256 symbols, root-raised-cosine pulses (roll-off 0.35), 3-tap complex Gaussian multipath normalized to unit gain, carriers 2000 Hz/2005 Hz with ± 5 Hz jitter, $T = 4096$ samples [14]), mixed with $\alpha_1 = \alpha$, $\alpha_2 = 1 - \alpha$, $\alpha \sim U(0.4, 0.6)$, plus AWGN. Throughout, SNR denotes the *mixture* SNR $\text{SNR}_{\text{mix}} = 10 \log_{10}(\mathbb{E}[\alpha_1 s_1 + \alpha_2 s_2]^2 / \mathbb{E}[n]^2)$. Since the sources are unit-power, the per-source SNR of source i is $\text{SNR}_i = \text{SNR}_{\text{mix}} + 10 \log_{10}(\alpha_i^2 / (\alpha_1^2 + \alpha_2^2)) \approx \text{SNR}_{\text{mix}} - 3$ dB under near-equal mixing, the SIR $20 \log_{10}(\alpha_1 / \alpha_2)$ sweeps $[-3.5, 3.5]$ dB with the α draw, and the per-source label stored by the evaluation pipeline is this shared SNR_{mix} .

Modulations, protocols, pools. Modulations split into known $\mathcal{K} = \{\text{BPSK}, \text{QPSK}, \text{8PSK}, \text{16QAM}\}$ (training only) and held-out $\mathcal{U} = \{\text{64QAM}, \pi/4\text{-DQPSK}, \text{MSK}, \text{OFDM-QPSK}\}$. The OOD label $O_i \in \{k, u\}$ attaches to *source* i (u iff its modulation is in $\mathcal{U}$); test protocols $kk/ku/uu$ type the

Manuscript prepared 2026. B. Nong is with Tianfu College of Southwestern University of Finance and Economics, Mianyang 621000, China (e-mail: nongbin@tfswufe.edu.cn). W. Fu is with the School of Telecommunications Engineering, Xidian University, Xi'an 710071, China. Z. Jiang is with the School of Health Science and Engineering, University of Shanghai for Science and Technology, Shanghai 200093, China.

mixture by its number of unknown sources (deterministic seed 99999; 192 kk, 192 ku, 96 uu mixtures per bin of the grid $\{-10, -5, \dots, 20\}$ dB), but detection is scored per source—a ku mixture contributes one known and one unknown sample, the basis of the same-mixture control (Sec. IV). The *known pool* holds both slots of the kk mixtures and the *unknown pool* the OOD slot of each ku mixture plus both slots of uu (2688 sources each; 384 per SNR bin). The separator is a lightweight complex-valued CNN with complex squeeze-and-excitation (C-SE) attention (~ 69 K parameters) plus a shared per-source modulation head (64-d embedding z , 4 logits f), trained with permutation-invariant SI-SDR + cross-entropy [16], [17] on known modulations only (closed-set kk reference: SI-SDR -0.89 dB pooled, truth-anchored accuracy 0.52—an interference-limited regime, SIR ≈ 0 dB).

Scorers and leakage control. Six post-hoc OOD scorers (larger = more OOD) are applied per source: Energy $-\log \sum_k e^{f_k}$ [7], MSP [5], ODIN [6], Mahalanobis [8], nearest-prototype distance, and a VOS-inspired (training-free, post-hoc) virtual-outlier distance [9], all at literature defaults fixed *a priori* (ODIN $T=1000$, $\varepsilon=0.005$; Mahalanobis shared covariance, shrinkage 0.1; VOS-inspired scale 2); every fitted quantity (prototypes, covariance, virtual outliers, thresholds) is estimated on a disjoint reference pool (seed 88888, kk protocol), and **no unknown-class sample is used for any fitting, selection, or calibration step**—the routing boundary (0 dB) is a development-time choice, frozen from the original study’s validation-profile crossover before any corrected-test evaluation, and LOMO boundaries on a pseudo-ODD validation split (seed 77777). The rejection threshold τ is the 95th percentile of known-pool reference scores (5% false-rejection target). The a-priori *SNR-routed* score uses Energy at $\text{SNR} \leq 0$ dB and Prototype otherwise. We call a detector *deployable* only if every constituent—scorer, score orientation, routing rule, threshold—is fixed without unknown-class or test data. Four AUROC aggregations are distinguished throughout: *per-SNR* (within one bin), *pair-weighted* (bin weights $n_k n_u$), *pooled* (all bins concatenated), and *same-mixture* (the two slots within each ku mixture).

III. THE PITFALL

Mechanism and formalization. Index the test mixtures $m = 1, \dots, N$ in SNR-block order. The evaluation stores per-source scores q_j stacked *tile-wise* and per-source SNR labels ℓ_j written *interleaved*— $\text{np.repeat}([a, b], 2) = [a, a, b, b]$ duplicates each entry, $\text{np.tile}([a, b], 2) = [a, b, a, b]$ repeats the block—so score j belongs to mixture $\sigma_q(j) = ((j-1) \bmod N) + 1$ but is paired with the label of mixture $\sigma_\ell(j) = \lceil j/2 \rceil$: an indexing-permutation mismatch $\sigma_q \neq \sigma_\ell$ (Fig. 2). Binning by ℓ assigns to stored bin b known scores whose *true* SNRs differ from b wherever the permutations disagree; the exactly labeled unknown pool is unaffected, so each per-bin AUROC measures the score’s *SNR separability*, not OOD separability, with a magnitude set by the deterministic label–truth SNR gap (up to 15 dB here). A one-line audit exposes the corruption: binning the known pool’s stored *modulation* labels by the stored SNR labels gives impossible per-bin class counts (114/96/100/74 vs the exact 96); tile labels recover 96/96/96/96.

A conditional proposition. This failure instantiates a general confound whose conditions can be stated exactly.

Proposition 1. *Let S be an OOD score that is conditioned on the true SNR C but carries no OOD information beyond it, $S \perp O \mid C$ for the OOD label $O \in \{k, u\}$, and whose family $\{p(s \mid c)\}$ is stochastically ordered in c . If the stored labels assign to bin b a known pool whose true-SNR distribution $P(C_k \mid \ell_k=b)$ has mass at $c \neq b$, while the exactly labeled unknown pool sits at b , then the per-bin AUROC $A_b = P(S_u > S_k \mid \ell_u=\ell_k=b) \neq \frac{1}{2}$, with sign set by $b - \mathbb{E}[C_k \mid \ell_k=b]$.*

Sketch. Marginalizing the corrupted known labels, $A_b = \int P(S_u > S_k \mid c_u=b, c_k) dP(c_k \mid \ell_k=b)$; under $S \perp O \mid C$ the integrand equals $\frac{1}{2}$ at $c_k = b$ and moves monotonically with $b - c_k$ by stochastic ordering, so any mass away from b biases A_b . $\square$

Corollary 1 (Gaussian drift). *If $S \mid C=c \sim \mathcal{N}(\mu(c), \sigma^2)$ with μ increasing, then $A_b = \mathbb{E}_{c \sim P(C_k \mid \ell_k=b)} [\Phi((\mu(b) - \mu(c))/(\sqrt{2}\sigma))]$; to first order $A_b - \frac{1}{2} \propto \mu(b) - \mathbb{E}[\mu(C_k) \mid \ell_k=b]$: the per-bin bias is the score’s drift across the label–truth gap.*

A domain-free instance. The confound needs no separator, signal, or detector (script `toy_gaussian_confound.py`): $S = C + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 7^2)$, on the same grid, carries *no* OOD information by construction; correct labels give $A_b = 0.500$ at every bin, as Corollary 1 requires, while the repeat-vs-tile layout alone yields per-bin AUROCs 0.11–0.89 and reversed labels 0.001–0.999 (analytic, MC-verified), both weighted averages staying ≈ 0.50 .

Generality: permutation study. Re-binning the *corrected* scores with synthetic label permutations π over the known pool (5-seed mean weighted AUROC of the a-priori routed score): $\pi_{\text{identity}} = 0.508$; π_{repeat} (the original bug, isolated on clean pools) = 0.619; $\pi_{\text{reverse}} = 0.746$; ten uniform *random* permutations = 0.632 ± 0.005 ; block-structure preserving shifts are exact no-ops (0.508); per-bin AUROCs reach 0.85 under π_{reverse} . Two confounds combine: mismatched known bins separate by their true-SNR composition (exposing opposite monotone SNR gradients for Energy and Prototype), and the fixed a-priori routing rule harvests both gradients by construction. Notably, single-scorer weighted averages stay near 0.48–0.51 under every π —the average looks innocent while the per-bin structure is entirely spurious (the toy makes the same point analytically), which is how the bug presented as *complementarity*. Label–index alignment is therefore not an implementation detail but a confound-control condition for conditioned evaluation.

Why it survives the standard checks. The misalignment is a deterministic function of the data layout, identical for every seed: multi-seed averaging *confirms* it (± 0.031); significance tests pass precisely because the SNR-mismatch signal is real—what is wrong is the bins’ interpretation; robustness perturbations act on the signals, not the label geometry; and the pattern is mechanistically narratable (“confidence inverts at high SNR”), with a full supporting narrative (Fig. 1a). The critical check is label–score alignment against regenerated data. The bug surfaced only when a blind SNR estimator (median-eigenvalue subspace, 0.51 dB RMSE; M2M4 inad-

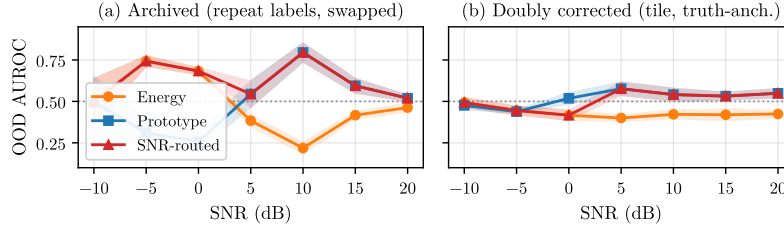


Fig. 1. Per-SNR OOD AUROC (5-seed mean $\pm$ std) under (a) the archived pipeline (repeat SNR labels, swapped pool membership) and (b) the doubly corrected evaluation (tile labels, truth-anchored). The apparent complementarity (Energy at low SNR, Prototype at high) is induced by the label misalignments; corrected, the routed profile is flat near chance (dotted).

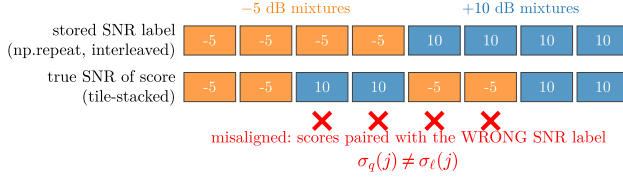


Fig. 2. The misalignment. Scores are stacked tile-wise (all slot-1 entries, then all slot-2), so their true SNR sequence alternates blocks; the stored labels repeat each mixture's SNR interleaved. Entries past the slot boundary (red crosses) are paired with the wrong SNR (up to 15 dB): $\sigma_q(j) \neq \sigma_l(j)$ (Sec. III).

equate here [13]) replaced the router's oracle SNR and the pairing check failed.

IV. CORRECTED PROTOCOL AND RESULTS

Protocol. (i) **Verified label alignment:** every dumped label array is checked against regenerated data (sequence equality, per-bin counts, per-bin label multisets; one-click script `audit_labels.py`). (ii) **Strict separation:** training, reference (seed 88888), and test (seed 99999) never mix; all fitted OOD quantities come from the reference pool. (iii) **Unknown-free calibration:** τ is the 95th percentile of known-pool reference scores; no unknown-class data is used for any fitted quantity, and the routing boundary is frozen from development (Sec. II). (iv) **Operating-point metrics**—Det@ τ , FPR95 (known-pool rejection rate at 95% unknown detection), OSCR [10], joint known/unknown accuracy—alongside per-SNR and pooled AUROC. (v) **Protocol decomposition:** pooled AUROC could confound modulation novelty with mixture composition (the known pool draws only from kk mixtures); we report protocol-wise AUROCs and a *same-mixture* comparison within each ku mixture, removing the confound by construction (uu has no known slot). (vi) **Independent test realizations:** beyond the deterministic test seed 99999, the full protocol is repeated on three further test sets (seeds 100001–100003). Seed stds describe training-run variability; bootstrap CIs test-sample variability.

A second misalignment, exposed by the same-mixture control. The same-mixture comparison required by protocol item (v) contradicted the pooled corrected numbers; the audit traced it to a second label misalignment in the dump stage (Fig. 3): permutation-invariant (PIT) alignment anchored the waveforms, embeddings, and logits to the true sources but stored

TABLE I
WEIGHTED-AVERAGE PER-SNR AUROC, MISLABELED VS CORRECTED LABELS (5 SEEDS, MEAN $\pm$ SEED STD); CORRECTED POOLED AUROC ON THE CONCATENATED 5-SEED SAMPLES (13440 + 13440) WITH ITS OWN 95% BOOTSTRAP CI; Δ = AUROC – 0.5; EVERY SCORER DIFFERS FROM CHANCE ($p \leq 0.01$, TWO-SIDED LABEL-PERMUTATION TEST, $B = 10^4$). CORRECTED COLUMNS ARE DOUBLY CORRECTED (TILE SNR LABELS, TRUTH-ANCHORED MEMBERSHIP AND REFERENCE FIT); THE MISLABELED COLUMN REPRODUCES THE ARCHIVED BUGGY DUMPS; VOS-INSPIRED SCORES COINCIDE WITH PROTOTYPE'S (SAME EMBEDDING GEOMETRY) AND ARE OMITTED. THE ORACLE'S MARGIN OVER ROUTED (0.042) SPLITS INTO MAX-SELECTION OPTIMISM (NULL 0.518 [0.513, 0.523]) PLUS A SMALL GENUINE MARGIN ($p = 5 \times 10^{-4}$ VS THE NULL).

Scorer	Corrected pooled [95% CI]		
	Buggy wavg	Corrected wavg	Δ
Energy	0.487 $\pm$ 0.032	0.431 $\pm$ 0.023	0.450 [0.443, 0.456], –0.050
MSP	0.435 $\pm$ 0.012	0.394 $\pm$ 0.007	0.405 [0.398, 0.411], –0.095
ODIN	0.495 $\pm$ 0.030	0.435 $\pm$ 0.021	0.460 [0.453, 0.466], –0.040
Mahalanobis	0.526 $\pm$ 0.043	0.529 $\pm$ 0.056	0.525 [0.517, 0.531], +0.025
Prototype	0.502 $\pm$ 0.008	0.518 $\pm$ 0.022	0.509 [0.502, 0.516], +0.009
SNR-routed	0.625 $\pm$ 0.031	0.508 $\pm$ 0.026	—
Oracle (6, post-hoc)	0.681 $\pm$ 0.039	0.550 $\pm$ 0.041	—

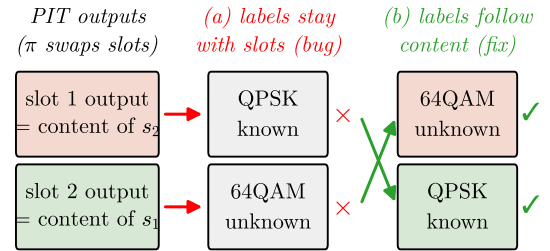


Fig. 3. The second misalignment (one ku mixture; PIT permutation π swaps the slots; s_1 QPSK known, s_2 64QAM unknown). PIT anchors output content to the true sources; the dump stored labels by slot (a); truth-anchored labels follow content (b).

the per-source *labels* (modulation indices and OOD flags) by output slot, contaminating about half of the ku-derived pools (swap rate ≈ 0.50 at SIR ≈ 0 dB) and attenuating every ku-dependent AUROC toward chance; the kk/uu pools, the reference pool, and the SNR labels are unaffected. The failure class is Proposition 1's—deterministic label misalignment—with the opposite sign: it *erases* structure. **All corrected numbers below are truth-anchored.**

Corrected results. Under verified, truth-anchored labels no scorer provides practically useful *pooled* discrimination (Fig. 1b; Table I): the routed ensemble reaches 0.508 ± 0.026

TABLE II
AUDIT MAP: CONTROL, CONFOUND EXCLUDED, OUTCOME (CORRECTED PROTOCOL).

Control	Confound excluded	Outcome
Label regeneration audit	label-score misalignment	archived dumps fail
PIT anchoring audit	source-label swap	closed-set acc. 0.41→0.52
Same-mixture contrast	mixture composition	margin 0.54
Bin-wise permutation null	post-hoc selection	oracle excess 0.032
LOMO; no-C-SE backbone	unknown set; architecture	no crossover; same

and stays flat across three further independent test sets (routed 0.502–0.515). The deviations from chance are structured: the logit scorers sit *below* chance (MSP $\Delta = -0.095$; Energy $\Delta = -0.050$; permutation $p = 10^{-4}$) while the embedding scorers sit marginally above (Mahalanobis $\Delta = +0.025$; Prototype/VOS $\Delta = +0.009$), driven by far-ODD classes and high-SNR bins (below). The post-hoc oracle (0.550) exceeds its correlation-preserving permutation null (0.518, $p = 5 \times 10^{-4}$)—a selection bound, not a deployable score (Table I). The intervention battery leaves the corrected routed score near chance: SIR/timing/CFO 0.488–0.504, carrier gaps ≤ 100 Hz 0.503–0.513 (500 Hz: 0.449—large separation breaks the co-frequency embedding), center loss ≈ 0.50 , embedding width 0.42–0.54; simulated SNR-estimation noise ($\sigma \leq 6$ dB) leaves routing unchanged. LOMO splits (boundary selected only on the pseudo-ODD validation split) give routed 0.482 ± 0.032 , and no split finds a genuine crossover (6/12 fall back to 0 dB, 6/12 to all-Energy). A second backbone without the C-SE blocks shows the same structure (routed 0.529 ± 0.021 , Mahalanobis 0.61).

Score orientation. Distinguish *no information* ($A \approx \frac{1}{2}$) from *inverted information* ($A < \frac{1}{2}$; AUROC($-S$) = $1 - \text{AUROC}(S)$), so the sign-flipped logit scorers reach 0.55–0.60—anti-correlated information, not none) and from a genuine but *non-deployable* margin. Every scorer is reported in its fixed literature orientation, frozen a priori: post-hoc sign choice would be the test-set selection the permutation null calibrates, and a validation-based rule cannot exist—validation has no unknown-class samples.

Protocol decomposition. Pool construction is not the driver (Table IV): the cross-mixture contrasts stay near chance, while the *same-mixture* contrast exposes a real but modest margin for the embedding scorers (Prototype/VOS 0.54, reaching 0.60–0.65 at SNR ≥ 5 dB) and a strong *inversion* for the logit scorers (Energy 0.33).

Operating point. At the calibrated single threshold (Table III) the detector rejects 6.4% of unknowns at 4.9% false rejection—near-useless, confirmed by FPR95 (0.947), OSCR (0.273) and joint accuracy (0.278). Per SNR, Det@ τ is *exactly* 0 at ≤ 0 dB and rises to 0.16 at 20 dB (false rejection $0 \rightarrow 0.126$): the threshold tracks the known pool’s SNR-driven drift and the unknown pool drifts identically—the failure is distributional, not a threshold choice.

Near vs far OOD. Per class, embedding scorers rank far-ODD families above near-ODD (0.54–0.58 on MSK/OFDM-QPSK vs 0.46–0.50 on 64QAM/ $\pi/4$ -DQPSK), while logit scorers *invert* the ordering (MSP 0.609 on $\pi/4$ -DQPSK vs 0.216 on OFDM-QPSK): the near/far intuition does not trans-

TABLE III
OPERATING-POINT METRICS OF THE ROUTED SCORE AT THE CALIBRATED 5%-FRR THRESHOLD τ (TRUTH-ANCHORED LABELS, 5 SEEDS, MEAN $\pm$ STD). A: SPLIT-HALF REFERENCE; B: DISJOINT REFERENCE POOL (SEED 88888), REFITTED PROTOTYPES.

Metric	Variant A	Variant B
Det@ τ (unknown detection)	0.066 $\pm$ 0.040	0.064 $\pm$ 0.040
False-rejection rate (target 0.05)	0.052 $\pm$ 0.000	0.049 $\pm$ 0.002
FPR95	0.947 $\pm$ 0.009	0.947 $\pm$ 0.008
OSCR [10]	0.272 $\pm$ 0.035	0.273 $\pm$ 0.034
Joint known/unknown accuracy	0.206 $\pm$ 0.042	0.278 $\pm$ 0.045

TABLE IV
PROTOCOL-DECOMPOSED WEIGHTED-AVERAGE AUROC (TRUTH-ANCHORED, 5-SEED MEAN); SAME-MIXTURE ROW: THE TWO SLOTS WITHIN EACH KU MIXTURE. ODIN/VOS BEHAVE LIKE ENERGY/PROTOTYPE.

Contrast	Ener.	MSP	Mah.	Prot.	Rout.
K(kk) vs U(ku)	0.393	0.373	0.526	0.524	0.506
K(kk) vs U(uu)	0.470	0.415	0.532	0.512	0.509
K(ku) vs U(ku), same mix.	0.334	0.347	0.517	0.543	0.523

fer to post-hoc scoring of separated sources.

Attribution. (1) Separation quality is class-independent (SI-SDR of known vs unknown sources -0.89 vs -1.00 dB pooled). (2) The margin tracks the weaker source’s separability: it peaks near equal power (0.63 at SIR 0/+5 dB, 10 dB mixture SNR) and degrades when either source dominates (≤ 0.55 at $|\text{SIR}| \geq 10$ dB). (3) A partial margin exists without interference (clean single-source probe: Mahalanobis 0.660 ± 0.046 ; logit scorers at or below chance). The exploitable margin *on separated sources* is thus real but modest, visible only against a same-mixture reference, and bounded by the weaker source’s residual interference; the logit pathway never carries one. Scope: the evaluated synthetic benchmark and protocol; unknown-aware open-set modulation recognition [4] is a separate, harder problem.

V. CONCLUSION

A repeat-vs-tile labeling mismatch manufactured an entire SNR-conditioned phenomenon that survived every randomness-oriented check; a PIT label swap did the opposite, erasing a genuine same-mixture margin. The failure is a general indexing confound (Corollary 1), not a one-off bug. Doubly corrected, pooled post-hoc OOD detection from separation embeddings remains non-deployable here; a modest same-mixture embedding margin survives as an honest starting point. We recommend: (1) *align labels against regenerated data* (audit_labels.py: sequence, per-bin counts, multisets) and *audit label anchoring after every permutation step* (closed-set accuracy is the zero-cost canary); (2) *separate train/reference/test strictly*, calibrate unknown-free, decompose pooled metrics by protocol; (3) *calibrate post-hoc selection against a permutation null*. Code and dumps: https://github.com/BinNong/single_channel_source_seperation/tree/paper3-spl/paper3_open_set.

REFERENCES

- [1] X. Hou and Y. Gao, "Single-channel blind separation of co-frequency signals based on convolutional network," *Digit. Signal Process.*, vol. 129, p. 103654, 2022.
- [2] P. Guo, M. Yu, L. Shen, Z. Lin, K. An, and J. Wang, "Single-channel blind source separation in wireless communications: a complex-domain deep learning approach," *IEEE Wireless Commun. Lett.*, vol. 13, no. 6, pp. 1645–1649, 2024.
- [3] W. J. Scheirer, A. de Rezende Rocha, A. Sapkota, and T. E. Boulton, "Toward open set recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 7, pp. 1757–1772, 2013.
- [4] T. Li, Z. Wen, Y. Long, Z. Hong, S. Zheng, and L. Shao, "The importance of expert knowledge for automatic modulation open set recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 11, pp. 13730–13748, 2023.
- [5] D. Hendrycks and K. Gimpel, "A baseline for detecting misclassified and out-of-distribution examples in neural networks," in *Proc. ICLR*, 2017.
- [6] S. Liang, Y. Li, and R. Srikant, "Enhancing the reliability of out-of-distribution image detection in neural networks," in *Proc. ICLR*, 2018.
- [7] W. Liu, X. Wang, J. D. Owens, and Y. Li, "Energy-based out-of-distribution detection," in *Proc. NeurIPS*, 2020.
- [8] K. Lee, K. Lee, H. Lee, and J. Shin, "A simple unified framework for detecting out-of-distribution samples and adversarial attacks," in *Proc. NeurIPS*, 2018.
- [9] X. Du, Z. Wang, M. Cai, and Y. Li, "VOS: Learning what you don't know by virtual outlier synthesis," in *Proc. ICLR*, 2022.
- [10] A. Dhamija, M. Günther, and T. Boulton, "Reducing network agnostophobia," in *Proc. NeurIPS*, 2018.
- [11] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100804, 2023.
- [12] K. Musgrave, S. Belongie, and S.-N. Lim, "A metric learning reality check," in *Proc. ECCV*, 2020, pp. 681–699.
- [13] D. R. Pauluzzi and N. C. Beaulieu, "A comparison of SNR estimation techniques for the AWGN channel," *IEEE Trans. Commun.*, vol. 48, no. 10, pp. 1681–1691, 2000.
- [14] J. G. Proakis and M. Salehi, *Digital Communications*, 5th ed., McGraw-Hill, 2008.
- [15] T. O'Shea and J. Hoydis, "An introduction to deep learning for the physical layer," *IEEE Trans. Cogn. Commun. Netw.*, vol. 3, no. 4, pp. 563–575, 2017.
- [16] M. Kolbæk, D. Yu, Z.-H. Tan, and J. Jensen, "Multitalker speech separation with utterance-level permutation invariant training of deep recurrent neural networks," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 25, no. 10, pp. 1901–1913, 2017.
- [17] J. Le Roux, S. Wisdom, H. Erdogan, and J. R. Hershey, "SDR—half-baked or well done?," in *Proc. ICASSP*, 2019, pp. 626–630.